

Demonstration of AWS Config Custom Compliance Rule Using Lambda

Many organizations require all EC2 instances to have mandatory tags such as Owner, Environment, and Project for cost allocation, governance, and operational management.

This project automatically identifies EC2 instances that do not meet these tagging standards.

Step-1: Login to your IAM user account and create an EC2 instance

The first screenshot shows the 'Launch an instance' page. A red box highlights the 'Name and tags' section, which includes fields for 'Name' and 'Owner', and a 'Tags' section with a 'Add new tag' button. The 'Application and OS Images' section shows the 'Quick Start' tab with various operating system options like Amazon Linux, macOS, Ubuntu, Windows, Red Hat, SUSE Linux, and Debian.

The second screenshot shows the 'Amazon Machine Image (AMI)' selection page. A red box highlights the 'Instance type' section, which shows the 't2.micro' instance type. The 'Key pair (login)' section shows a key pair named 'test_keypair'.

The third screenshot shows the 'Network settings' page. A red box highlights the 'Firewall (security group)' section, which shows a new security group named 'launch-wizard-4' with the following rules:

- ☒ Allow SSH traffic from 192.168.1.10/24
- ☐ Allow HTTPS traffic from the Internet
- ☐ Allow HTTP traffic from the Internet

The 'Configure storage' section shows a volume of 8 GB, gp3 type, with 'Root volume, 3000 IOPS, Not encrypted'.

Ste-2: Configure AWS Config and do not attach any AWS managed rules.

Recording method

Recording strategy

☒ All resource types with customizable overrides
This strategy records all resource types in the AWS account, except for resource types that are not supported by the recording method you selected. You can specify which resource types to record by using the Override settings section.

☐ Specific resource types
This strategy records only the resource types that you specify. You can specify which resource types to record by using the Override settings section.

Default settings

Recording frequency

☒ Continuous recording
This strategy records configuration changes continuously, unless a change occurs. This strategy is recommended for most use cases.

☐ Scheduled recording
This strategy records configuration changes at a specific time and frequency. You can specify the recording frequency by using the Override settings section.

Override settings

Recording frequency to override (Hz)

(Default: 1 Hz)

Resource types to override (IAM)

☒ All globally recorded IAM resource types
This strategy records all resource types in the AWS account, except for resource types that are not supported by the recording method you selected. You can specify which resource types to record by using the Override settings section.

☐ Specific resource types
This strategy records only the resource types that you specify. You can specify which resource types to record by using the Override settings section.

Data governance

IAM role for AWS Config

☒ Create AWS Config service-linked role
This strategy creates a new IAM role for AWS Config. This role is used to record configuration changes and notifications to an Amazon SNS topic.

☐ Choose a role from your account
Choose an IAM role from one of your pre-existing roles and permission policies.

Delivery channel

Amazon S3 bucket

☒ Create a bucket
☐ Choose a bucket from your account
☐ Choose a bucket from another account

Ensure appropriate permissions are available in this S3 bucket's policy. [Learn more](#)

S3 Bucket name (required)

/AWSLogs/559050212593/Config/ap-south-1

Amazon SNS topic

☐ Stream configuration changes and notifications to an Amazon SNS topic.
If you choose email as the notification endpoint for your SNS topic, this can cause a high volume of email. [Learn more](#)

[Cancel](#) [Next](#)

Step-3: Create an IAM role that allows Lambda to access EC2 information and send compliance results to AWS Config.

Select trusted entity

Trusted entity type

☒ AWS service
Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

☐ AWS account
Allow an AWS account to perform actions in this account.

☐ With identity
Allow an AWS account to perform actions in this account.

☐ Custom trust policy
Create a custom trust policy to enable others to perform actions in this account.

Use case

Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case

Choose a use case for the specified service.

Use case

☒ Lambda
Allow Lambda functions to call AWS services on your behalf.

☐ AWS service role for Lambda
Allow Lambda functions to call AWS services on your behalf.

[Cancel](#) [Next](#)

Role name

Description

Step 1: Select trusted entities

Trust policy

```

{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": "ec2:*",
      "Resource": "*"
    },
    {
      "Effect": "Allow",
      "Action": "logs:*",
      "Resource": "*"
    },
    {
      "Effect": "Allow",
      "Action": "sns:*",
      "Resource": "*"
    }
  ]
}

```

Step 2: Add permissions

Permissions policy summary

Permissions policy	Type	Attached to
AWS managed	AWS managed	Permissions policy
AWS managed	AWS managed	Permissions policy

Step 3: Add tags

Add tags - optional

Tags are key-value pairs that you can use to identify resources in your AWS account. The tags associated with this resource.

Step-4: Create lambda function and attach the IAM role you have created by choosing custom execution role in additional settings and save it.

Create function info

Choose one of the following options to create your function.

- Author from scratch** Start with a simple Hello World example.
- Use a blueprint** Build a Lambda application from sample code and configuration presets for common use cases.
- Container image** Select a container image to display for your function.

Basic information

Function name
Enter a name that describes the purpose of your function.

Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).

Runtime info
Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.

Permissions
By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. To use a different role, choose **Custom execution role** under **More settings**.

Custom settings info

Durable execution - new ? Build resilient, stateful applications with automatic failure recovery.

EC2 capacity provider - new ? Run Lambda on your preferred EC2 instance types instead of Lambda's serverless compute (default).

Additional settings
Configure your function architecture, execution role, HTTPS endpoints, tenant isolation mode, VPC, code signing, KMS key, and tags.

General info

ARM64 architecture info
Use ARM64 instead of x86_64 (default).

Custom execution role info
Use custom execution role instead of new role with basic Lambda permissions (default).

Networking info

Function URL info
Assign HTTPS endpoints.

Tenant isolation mode - new ? Guarantee separate execution environments for each tenant invoking the function.

VPC info
Connect to a VPC to securely access private resources.

Security & governance info

Configure custom execution role

Choose an existing execution role or create a new one.

Execution role
Access other AWS services and resources your function needs an IAM role. Choose a role with the right permissions for your function.

[Create new role](#) [View role details in IAM](#) [Clear](#) [Save](#)

Step-5: In IAM roles add an inline policy for config so that Lambda can put evaluation result in aws config and also update the lambda function by adding code init.

Review and save info

Review the permissions, specify details, and tags.

Permissions defined in this policy info

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, group, or role), attach a policy to it.

Allow [1 of 470 services] [Show remaining 469 services](#)

Service	Access level	Resource	Request condition
Config	Limited Write	All resources	None

EC2TagComplianceCheck

Function overview info

[Blueprint](#) [Template](#)

EC2TagComplianceCheck

[Add trigger](#) [Add destination](#)

Function ARN
[arn:aws:lambda:ap-south-1:53905212595:function:EC2TagComplianceCheck](#)

Description
-

Last modified
7 minutes ago

Code source info

[Open in Visual Studio Code](#) [Update](#)

```

1 import boto3
2
3 config = boto3.client('config')
4
5 REQUIRED_TAGS = ["Owner", "Environment", "Project"]
6
7 def lambda_handler(event, context):

```

Step-6: create aws config custom rule so that whenever ec2 instance tags created or modified, aws config will automatically evaluates compliance.

Configure rule
Customize any of the following fields

Details

Name
A unique name for the rule. 128 characters max. No special characters or spaces.
ec2-required-tags-check

Description - optional
Describe what the rule evaluates and how to fix resources that don't comply.
Checks whether EC2 instances contain Owner, Environment and Project tags.

AWS Lambda Function ARN
The AWS Lambda function that evaluates whether your AWS resources comply with the rule. AWS Config invokes this function when the rule is triggered.
arn:aws:lambda:ap-south-1:559050212593:function:EC2TagComplianceCheck

Evaluation mode

☒ Turn on proactive evaluation
Enables evaluation of resources prior to provisioning.

☐ Turn on detective evaluation
Enables evaluation of resources which have been provisioned.

Trigger type
AWS Config evaluates resources when the trigger occurs.
☒ When configuration changes
Runs when there are changes to your specified AWS resources.
☐ Periodic
Runs on the frequency that you choose.

Scope of changes
Choose when evaluations will occur.

☐ All changes
When any resource recorded by AWS Config is created, changed, or deleted.

☒ Resources
When any resource that matches the specified type, or the type plus identifier, is created, changed, or deleted.

☐ Tags
When any resource with the specified tag is created, changed, or deleted.

Resources
This can only be triggered only when the recorded resources are created, edited, or deleted. Specify the resources to be tested by editing the settings scope.

Resource category
All resource categories

Resource type
Multiple selected

Resource identifier - optional
Enter resource identifier

Parameters
Rule parameters define attributes that your resources must adhere to for compliance with the rule. Example attributes include a required tag or a specified S3 bucket. Optional parameters that are not valid, such as mixing a key or a value, will not be saved.

Key
Key

Value
(optional)

[Remove](#)

[Add another row](#)

Step-7: After 3-5 min you can see ec2 instance we have created as non-compliant as we are missing two tags project and environment.

ec2-required-tags-check

Rule details

Description
Checks whether EC2 instances contain Owner, Environment and Project tags.

Config rule ARN
arn:aws:config:ap-south-1:559050212593:config-rule/config-rule-gpu-ty

Enabled evaluation mode
• DETECTIVE

Last successful detective evaluation
June 9, 2025 7:07 PM

Detective evaluation trigger type
• Overwritten configuration changes
• Configuration changes

Scope of changes
Resources

Resource types
EC2 Instance

Resources in scope

ID	Type	Status	Annotation	Compliance
i-010e7d32f56a2211	EC2 Instance	Noncompliant		Noncompliant

Step-8: Go to Instance created and select tags ->manage tags and add the project and environment tags then you can see instance as compliant.

Instances (1/1)

Test-Server

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4	Elastic IP	IPv6 IP
Test-Server	i-010e7d32f56a2211	Running	t2.micro	2/2 checks passed	ap-south-1b	ec2-15-201-228-178.ap...	15.201.228.178	-	-

i-010e7d32f56a2211 (Test-Server)

Tags

Key	Value
Owner	Sowmya
Name	Test-Server

Search

AB1+1

EC2

Instances

test_ubuntu

Manage tags

Manage tags

A tag is a named label that you assign to an AWS resource. You can use tags to help organize and identify your instances.

Key

Owner

X

Value - optional

Seamys

X

Remove

Q

Name

X

Q

Test-Server

X

Remove

Q

Environment

X

Q

DEV

X

Remove

Q

Project

X

Q

AWS-Config-Demo

X

Remove

Add new tag

You can add up to 50 more tags.

Cancel

Save

AWS Config

Rules

ec2-required-tags-check

Actions

ERR

Rule details

Description

Checks whether EC2 instances contain Owner, Environment and Project tags.

Enabled evaluation mode

DETECTIVE

Detective evaluation trigger type

Overloaded configuration changes

Configuration changes

Config rule ARN

arn:aws:config:ap-south-1:559850212593:config-rule/config-rule-gptuys

Last successful detective evaluation

June 9, 2026 7:15 PM

Scope of changes

Resources

Resource types

EC2 Instance

Resources in scope

All

View details

Remediate

ID	Type	Status	Annotation	Compliance
i-010e76529f56a2211	EC2 Instance	-	-	Compliant

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